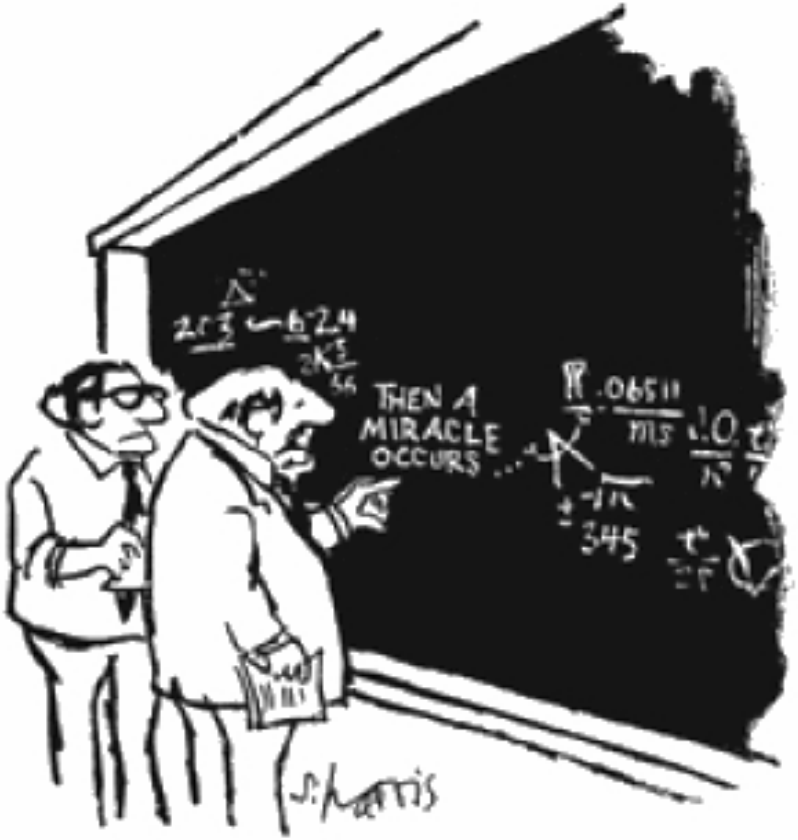
The Capital Asset Pricing Model



"I think you should be more explicit here in step two."

CAPM: Introduction

Capital Asset Pricing Model.

- The capital asset pricing model (CAPM) is used to determine a theoretically appropriate required rate of return of an asset, if that asset is to be added to an already well-diversified portfolio (market portfolio).
- The model takes into account the asset's sensitivity to market risk (also known as systematic risk or non-diversifiable risk), often represented by the quantity beta (β) in the financial industry, as well as the expected return of the market and the expected return of a theoretical risk-free asset.

A Preview of the CAPM

- In a nutshell, the CAPM says that

$$\overbrace{r_{\text{Apple}} - r_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Apple}} \times (r_{\text{Market}} - r_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Apple}}}^{\text{Idiosyncratic Risk}}$$

  where r_F is the risk-free rate and ϵ_{Apple} is a random shock to Apple's return (e.g. problems with an Apple supplier, Steve Jobs dies, the Justice Department figures out how to hack iPhones, etc.).
- **Important:** $E[\epsilon_{\text{Apple}}] = 0$.
- The CAPM says that

$$\overbrace{E[r_{\text{Apple}}] - r_F}^{\text{Apple Risk Premium}} = \beta_{\text{Apple}} \times \overbrace{(E[r_{\text{Market}}] - r_F)}^{\text{Equity Risk Premium}}$$

- Investors are not compensated for idiosyncratic risk.

Sensitivity to Systematic Risk: Beta

$$\overbrace{r_{\text{Apple}} - r_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Apple}} \times (r_{\text{Market}} - r_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Apple}}}^{\text{Idiosyncratic Risk}}$$

- *The beta of a security is the expected % change in its return given a 1% change in the return of the market portfolio.*
- General Mills has a beta of 0.4. People eat about the same amount of Cheerios regardless of the state of the economy.
- Sotheby's has a beta of 2.17. People buy fewer Picassos in recessions than in booms.

CAPM Beta: Examples

Walmart Inc. (WMT) Add to watchlist

NYSE - NYSE Delayed Price. Currency in USD

96.90 +1.07 (+1.12%) At close: 4:00PM EDT

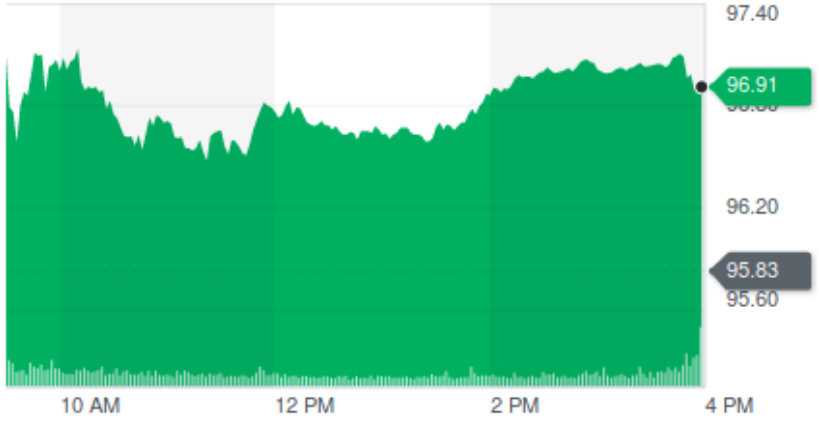
96.95 +0.05 (0.05%) After hours: 6:43PM EDT

Buy Sell

Summary | Chart | Conversations | Statistics | Profile | Financials | Options | Holders | Historical Data | Analysis | Sustainability

Previous Close	95.83	Market Cap	283.794B
Open	96.11	Beta	0.25
Bid	0.00 x 1100	PE Ratio (TTM)	55.43
Ask	0.00 x 3000	EPS (TTM)	1.75
Day's Range	96.09 - 97.19	Earnings Date	Nov 15, 2018
52 Week Range	77.50 - 109.98	Forward Dividend & Yield	2.08 (2.16%)
Volume	8,332,605	Ex-Dividend Date	2018-12-06
Avg. Volume	7,814,630	1y Target Est	104.00

1D 5D 1M 6M YTD 1Y 5Y Max Full screen



CAPM Beta: Examples

Tiffany & Co. (TIF) Add to watchlist

NYSE - NYSE Delayed Price. Currency in USD

124.32 +1.45 (+1.18%) At close: 4:02PM EDT

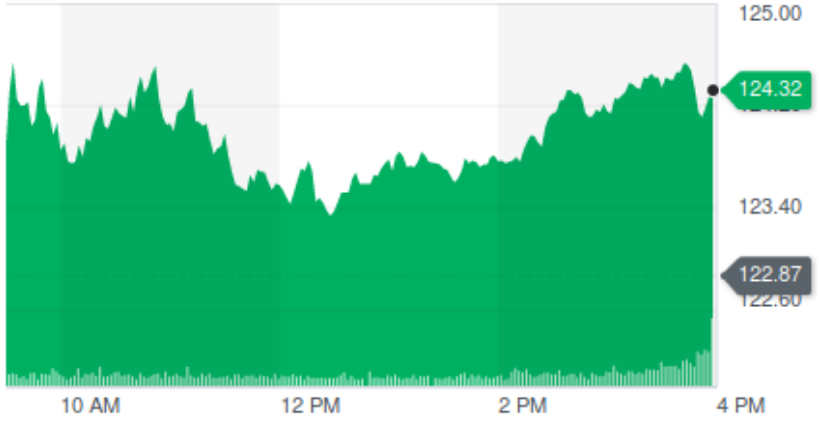
124.09 -0.23 (-0.19%) After hours: 4:26PM EDT

Buy Sell

Summary | Chart | Conversations | Statistics | Profile | Financials | Options | Holders | Historical Data | Analysis | Sustainability

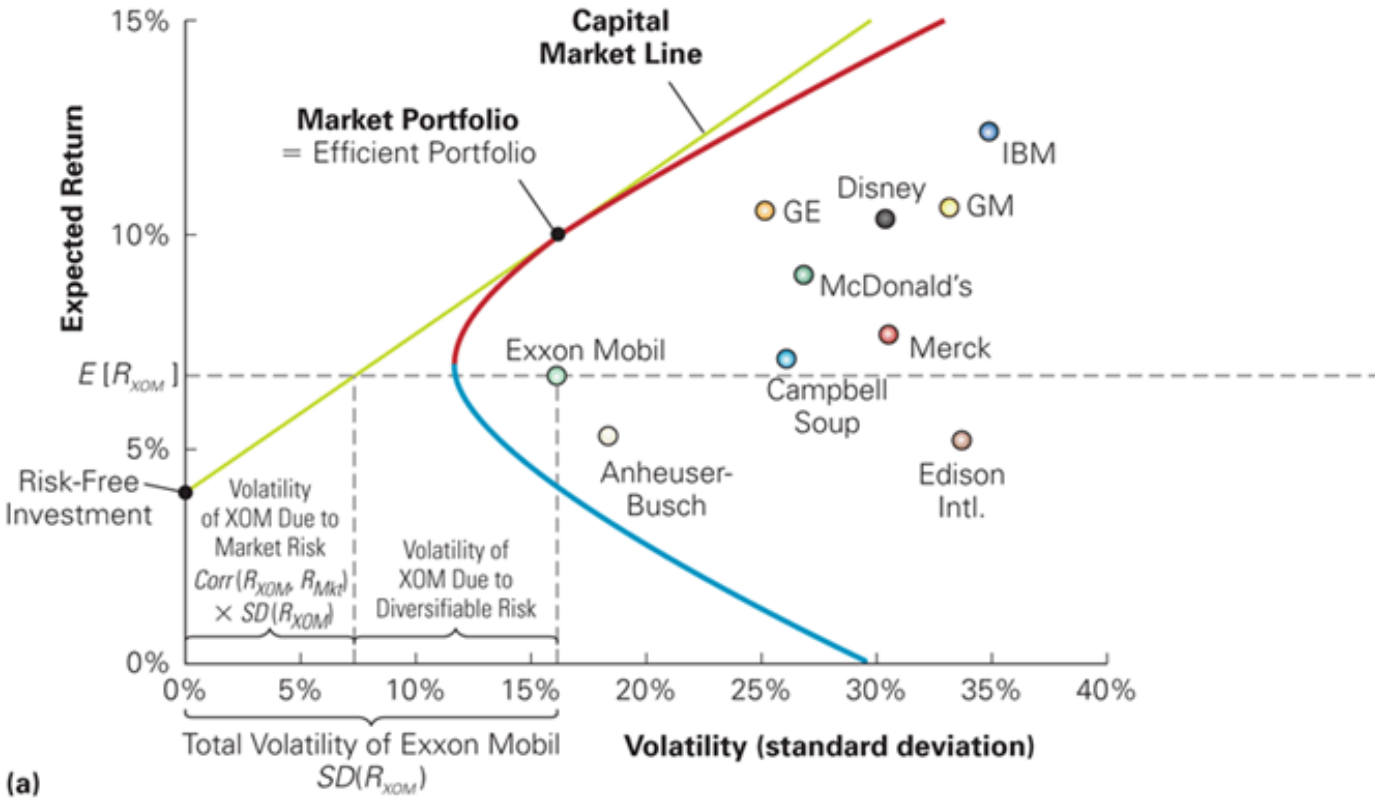
Previous Close	122.87	Market Cap	15.218B
Open	123.86	Beta	1.47
Bid	123.40 x 800	PE Ratio (TTM)	34.46
Ask	124.76 x 1400	EPS (TTM)	3.61
Day's Range	123.29 - 124.80	Earnings Date	Nov 27, 2018 - Dec 3, 2018
52 Week Range	86.15 - 141.64	Forward Dividend & Yield	2.20 (1.78%)
Volume	1,498,549	Ex-Dividend Date	2018-09-19
Avg. Volume	1,598,087	1y Target Est	140.50

1D 5D 1M 6M YTD 1Y 5Y Max Full screen



Important Result from Mean-Variance Frontier and CML

$$\overbrace{r_{\text{Apple}} - r_F}^{\text{Stock Return}} = \overbrace{\beta_{\text{Apple}} \times (r_{\text{Market}} - r_F)}^{\text{Systematic Risk}} + \overbrace{\epsilon_{\text{Apple}}}^{\text{Idiosyncratic Risk}}$$



Diversifiable and Undiversifiable Risk

- CAPM:

$$R_S - R_F = \beta_S (R_M - R_F) + \varepsilon_S$$

- Take the variance:

$$Var(R_S - R_F) = \beta^2 Var(R_M - R_F) + Var(\varepsilon_S)$$

- Remember, $Cov(R_M - R_F, \varepsilon_S) = 0$.
- ε_S is all of the random good/bad news during the day that doesn't have anything to do with whether the market is up/down.

CAPM: The Model

- CAPM states that security risk premiums (expected excess returns) will be proportional to beta:

$$E(R_S) = R_F + \beta_S[E(R_M) - R_F]$$

where beta is defined as:

$$\beta_S = \frac{cov(R_S, R_M)}{Var(R_M)}$$

- β_S can be interpreted as the exposure of stock S to the market risk. $E(R_M) - R_F$ is also know as the market premium.

Example

Assume that the correlation between Apple stock and the Market Portfolio is 0.4. The standard deviation of Apple is 36% and the standard deviation of the Market Portfolio is 12%. Also assume that the expected return of the market portfolio is 8% and the risk-free rate return in 3%. What is the beta of apple? What is the expected return of Apple under the CAPM?

CAPM and Realized Returns

- If CAPM holds, do realized returns always follow the rule below?

$$R_S = R_F + \beta_S \left(R_M - R_F\right)$$

- NO! If CAPM is true realized returns should follow:

$$R_S - R_F = \beta_S \left(R_M - R_F\right) + \varepsilon_S$$

  As long as $E(\varepsilon_S) = 0$ CAPM holds.
- What is CAPM is not true? Then we need to add an alpha (α):

$$R_S - R_F = \alpha_S + \beta_S \left(R_M - R_F\right) + \varepsilon_S$$

What if CAPM does NOT Hold? Alpha (α)

- **Alpha (α):** The difference between an asset's actual expected return and what something with its beta should return according to the CAPM is called the asset's **alpha**:

$$\alpha_i = \underbrace{E(R_i)}_{True\ \ Value} - \underbrace{[R_f + \beta_i (E(R_m) - R_f)]}_{CAPM}$$

- If CAPM holds then $\alpha = 0$, but if CAPM does NOT hold then $\alpha \neq 0$.

Example

Suppose that the expected market return is 14% and the risk-free rate is 6%. Consider a stock with a beta of 1.2. Suppose that you find that the stock has an average annual return of 18%. What is the alpha of this stock?

Example

Suppose that the risk-free rate is zero and the market return is r_M. Consider a stock with alpha $\alpha > 0$ and beta β. What arbitrage opportunity is available and how would you exploit it?

Multifactor Models

$$r_S - r_F = \beta_1 f_1 + \beta_2 f_2 + \cdots + \beta_n f_n + \epsilon_S. \tag{1}$$

- Fama and French (1992) proposed a three factor model:
  1. The market return minus the risk-free return
  2. A portfolio that longs small stocks and shorts large ones
  3. A portfolio that longs high book-to-market stocks and shorts low book-to-market ones.
- There is a cottage industry in finding new factors.
- Modern hedge funds may use dozens of factors.